AFF–1069

Sub. Code
4MZO1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Zoology

ANIMAL DIVERSITY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Father of Taxonomy.

2. Plasmid.

3. Cilia.

4. Mesoglia.

5. Edible oyster.

6. Trachia.

7. Retrogressive metamorphosis.

8. Swim bladder.

9. Metamerism.

10. Contour feathers.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) What are the salient features of animal classification?

Or

(b) Briefly describe about the symmetry.

12. (a) Write notes on helminthes parasites.

Or

(b) Briefly describe about the classification of Mollusca.

13. (a) Write note on biramous appendages of Arthropoda.

Or

(b) Why insects are more diverse than others?

14. (a) Write note on classification of Amphibians.

Or

(b) What are the general characters of Amphioxus?

15. (a) Describe about the salient features of reptiles.

Or

(b) What are the general characters of aves?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about the major divisions and sub divisions of animal kingdom.
17. Write about the life cycle of liver fluke
18. Explain about the water vascular system of Echinodermata.
19. Describe about the classification of mammals and their general characters.
20. Describe about the migration of fishes.

AFF–1070

Sub. Code
4MZO1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Zoology

BIOCHEMISTRY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Deamination.

2. Essential Amino acid.

3. Amylase.

4. Starch.

5. LDL.

6. Rancidity.

7. Prosthetic group.

8. Zymogens.

9. TSH

10. Adrenal gland.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Write short notes on the secondary structure of proteins.

Or

(b) Briefly describe about the method of isolation of proteins.

12. (a) Write note on gluconeogenesis.

Or

(b) Briefly describe about the formation of Acetyl Co A.

13. (a) Write short note on Ketogenesis and its regulations.

Or

(b) What are the role of RNA and how they are classified?

14. (a) Briefly describe about the properties of enzymes.

Or

(b) Write short note on the classification of enzymes.

15. (a) Briefly describe about the function of thyroid hormones.

Or

(b) What are the functions of fat soluble vitamins?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Write an essay about the isolation and characterization of protein.

17. Describe about the Glycolysis and its energetics.

18. Explain about the structure and function of DNA.

19. What are the types of vitamins and explain their functions.

20. Write notes on the biosynthesis of hormones.

AFF–1071

Sub. Code
4MZO1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Zoology

CELL AND MOLECULAR BIOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Lysosomes.

2. Peroxysomes.

3. Okazaki fragments.

4. Origin of replication.

5. Positive control.

6. Exons.

7. G-protein coupled receptors.

8. Gap junction.

9. Stem cell therapy.

10. Adult stem cell.

Part B (5 × 5 = 25)

Answer **all** questions choosing either (a) or (b).

11. (a) Write notes on Golgi complex.

Or

(b) Draw the ultra structure of chloroplast.

12. (a) Write short note on prokaryotic DNA replication.

Or

(b) Briefly describe about the DNA damage and repair.

13. (a) Explain about the transcription of DNA.

Or

(b) Write note on post translational modifications.

14. (a) Write note on signal transduction pathways.

Or

(b) Briefly describe about the cell adhesion and its role.

15. (a) Write note on type of scaffolds.

Or

(b) What are the ethical issues involved in the use of stem cells?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about the structure and function of Mitochondria.

17. Explain about DNA replication in Eukaryotes.

18. Explain the mechanism of positive and negative controls in operon.

19. Describe about the secondary messengers.

20. Write note on organ culture and stem cell therapy.

AFF–1072

Sub. Code
4MZO1E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Zoology

***Elective* — BIOSTATISTICS**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Frequency polygon.

2. Bar graph.

3. Mode.

4. Standard error.

5. ANOVA.

6. Poisson distribution.

7. P value.

8. Type I error.

9. '*t*' test.

10. Regression.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the characteristics of classification?

Or

(b) Explain about the different types of tables.

12. (a) Write short notes on range, standard deviation and variance.

Or

(b) Describe briefly about co-efficient of variation.

13. (a) Enumerate and briefly explain about the parameters and properties of binomial distribution.

Or

(b) Give an account on the applications of ANOVA.

14. (a) What is chi-square test? Explain its applications in biology.

Or

(b) Explain the following :

(i) Null-hypothesis

(ii) Level of significance and

(iii) Degrees of freedom.

15. (a) Define Correlation. What are the different types of correlation?

Or

(b) What is regression analysis? What are its biological properties?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. The following data is on the number of three species of snakes – Cobra, Viper and Krait in four districts of Tamil Nadu. Represent the data in multiple bar diagram, component bar diagram and percent bar diagram.

Districts	No. of Snakes		
	Cobra	Viper	Krait
Coimbatore	45	22	12
Erode	30	25	18
Salem	35	18	22
Dharmapuri	60	30	19

17. Find out the mean, median and mode for the number of eggs per nest of a species of bird – a discrete frequency distribution :

No. of eggs/nest	0	1	2	3	4	5
Frequency	5	8	12	12	12	7

18. Explain the one-way and two-way classification of ANOVA.

19. The following data on the weights in *g* of species of fishes before and after feeding a type of food, test whether a significant mean difference exists using student *t* test :

Before	17	17	18	21	22	17	23	23
After	18	17	20	24	23	15	25	22

20. Obtain the coefficient of correlation for the following data on the length (X in cm) and weight (Y in g) of fish:

X	5	7	3	1	9	12	8	3
Y	8	9	5	4	9	13	7	9

AFF–1073

Sub. Code
4MZO1E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Zoology

Elective – ANIMAL BEHAVIOUR

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Genetic influence on behaviour.

2. Annual rhythm.

3. Bird pollination.

4. Social relationship.

5. Group formation.

6. Climatic influence of behaviour.

7. Ethogram.

8. Ultimate causation.

9. Kin selection.

10. Operant conditioning.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the importance of animal behaviour studies?

Or

(b) Write note on physiological basis of behaviour.

12. (a) Describe about the migration of fishes?

Or

(b) Give a brief account on importance of migration.

13. (a) Write short note on process of socialization.

Or

(b) Briefly describe about the homing of animals.

14. (a) Write note on analysis of social behaviour.

Or

(b) How the physical environment affect the behaviour of animals?

15. (a) What is Hamilton's rule?

Or

(b) Write an account on animal cognition.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about the genetic differences in the behaviour of animals and behavioural disorders.

17. Give an account on migration of fishes and crustaceans and its importance.

18. Write an essay about behavioural characters for management practices.

19. What are the methods used in the analysis of social behaviour?

20. Write an essay about the biological clock.

———————

AFF–1075

Sub. Code
4MZO2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Zoology

GENETICS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Phenotype

2. Test cross

3. Histone

4. Barr bodies

5. Morgan

6. RAPD

7. Codon

8. Bacteriophage

9. Genetic bottle neck

10. Twin study

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short note on monohybrid cross.

Or

(b) Explain about the pedigree analysis.

12. (a) Write short note on mechanism of crossing over.

Or

(b) Give a brief account on point mutation.

13. (a) Write short note on tetrad analysis.

Or

(b) Briefly describe about QTL mapping.

14. (a) Write a short note on Lac operon.

Or

(b) Write brief account on sequential expression of genes.

15. (a) Write short note on founder principle.

Or

(b) Briefly discuss about the biological clock.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss about polygenic inheritance and multiple alleles.

17. Write an essay about linkage and crossing over.

18. Describe about the chromosome mapping methods.

19. Describe about the genetic regulation of development.

20. Explain about Hardy Weinberg Law and how it get affected during the genetic drift.

AFF–1076

Sub. Code
4MZO2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Zoology

IMMUNOLOGY AND MICROBIOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Antigen.

2. Natural Killer cells.

3. Immuno prophylaxis.

4. Vaccine.

5. Immuno precipitation.

6. AIDS.

7. Gram positive bacteria.

8. Pure culture.

9. Herpesviridae.

10. Pasteurization.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write note on cell mediated immune response.

Or

(b) Give an account on immuno deficiency diseases.

12. (a) Briefly describe about Western blotting.

Or

(b) Describe about innate and acquired immunity.

13. (a) Write short note on hybridoma technology.

Or

(b) Briefly describe about the Immunoflorescence microscope.

14. (a) Write note on bacterial growth curve.

Or

(b) Draw the ultra structure of bacteriophage.

15. (a) Describe about the strain improvement methods.

Or

(b) Give a brief account on mycotic infections.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe about primary and secondary lymphoid organs.

17. Write an essay about transplantation immunology.

18. What are culture media used for bacteria and viruses?

19. Write an essay about classification and identification of bacteria.

20. Describe about any four viral diseases.

AFF–1078

Sub. Code
4MZO2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Zoology

Elective – WILDLIFE CONSERVATION AND MANAGEMENT

(CBCS – 2014 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Genetic diversity

2. Biodiversity Hotspots

3. NBA

4. WWF

5. Buffer zone

6. Chipko movement

7. Population density

8. Red data book

9. *In situ* Conservation

10. Endangered species.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write short note on biodiversity.

Or

(b) Write short note on species diversity.

12. (a) Describe about hotspots in India.

Or

(b) Give an account on endangered species in India.

13. (a) Write short note on wildlife ecosystem management.

Or

(b) Briefly describe the habitat analysis and design.

14. (a) Give an account on rules and regulation of the Central Zoo Authority of India.

Or

(b) Give an account on wildlife legislation in India.

15. (a) Describe about the distribution of important Indian wild animals.

Or

(b) What are the funding agencies available for wildlife research?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Briefly explain about basic principles of formation of sanctuaries.

17. What are the major threats to biodiversity?

18. Give an account on prey predator relationship.

19. Write an essay on natural resources and their management for wildlife conservation.

20. What are the voluntary organizations involved in the conservation of wildlife?

AFF–1079

Sub. Code
4MZO3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

DEVELOPMENTAL BIOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Fertilization
2. Cleavage
3. Metamorphosis
4. Cell Motility
5. Organogenesis
6. Notochord
7. Organizer
8. Placenta
9. Cryopreservation
10. IVF

Part B (5 × 5 = 25)

Answer **all** questions, by choosing either (a) or (b).

11. (a) Explain the ultrastructure of Sperm.

Or

(b) Briefly describe the types of Symmetry.

12. (a) Explain about morphogenesis.

Or

(b) Briefly describe the types of eggs.

13. (a) Write short note on organogenesis.

Or

(b) Describe about cryopreservation.

14. (a) Write short note on types of mammalian placenta.

Or

(b) Write an account on nuclear transplantation.

15. (a) Describe about regeneration in planaria.

Or

(b) Describe about gastrulation of frog.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Briefly describe about the menstrual cycle.

17. Give an account on the development of Eye in chick.

18. Explain in detail about the metabolic and molecular changes during gastrulation.

19. What is cryopreservation and their application in modern embryology?

20. Explain about *in vitro* fertilization and their applications.

———————

AFF–1080

Sub. Code
4MZO3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

Core Course X – ECOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Natality.

2. Mutualism.

3. Ecological Niche.

4. Pyramid of energy.

5. Nitrification.

6. Respiration.

7. Mangroves.

8. Limnology.

9. UNEP.

10. Bioremediation.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) What are the difference between community and population?

Or

(b) Write short notes on the following :

(i) Commensalism

(ii) Parasitism and

(iii) Competition.

12. (a) What is an ecosystem? How does energy flow in an ecosystem?

Or

(b) Describe about the components of an ecosystem.

13. (a) What is the role of phosphorus cycle in circulation of nutrients in an ecosystem?

Or

(b) Explain the cycling of organic nutrients in an ecosystem.

14. (a) Write short notes on the adaptations of desert animals.

Or

(b) Briefly describe about the biological features of Mangroves.

15. (a) Discuss about the effect of climate change on living organisms.

Or

(b) Explain the role of microorganisms in bioremediation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Define the term ecological succession? Describe the process of ecological succession in a pond?

17. What are ecological pyramids? Describe various types of ecological pyramids giving suitable examples?

18. "Carbon cycle is essentially a perfect cycle" In your opinion, how far is this statement true?

19. What are natural resources? Why is the conservation of natural resources a must for mankind?

20. List out the major environmental laws applicable in India and also describe about the various organizations involved in environmental protection?

AFF–1081

Sub. Code
4MZO3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

EVOLUTION

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Darwinism.

2. Significance of Hugo de Vries' Mutation Theory.

3. Retrogressive Metamorphosis.

4. Palaeontology.

5. Temporal isolation.

6. Gametic incompatibility.

7. Colouration.

8. Darwin's finches.

9. Radiodating.

10. Founders effect.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Give a brief account on Neo Lamarkism.

Or

(b) Explain Wallace theory.

12. (a) Give vestigial organs with examples.

Or

(b) Explain atavism with examples.

13. (a) Describe Pre-mating isolating mechanisms.

Or

(b) What are all the sources of genetic variation?

14. (a) Write short notes on struggle for existance.

Or

(b) Explain co-evolution.

15. (a) Describe social evolution.

Or

(b) Write short notes on Molecular clock.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain Big Bang theory in detail.

17. Write an essay on physiological and biochemical Evidences of evolution with examples.

18. Describe genetic variation and factors affecting genetic variation.

19. Explain identification, causes and effects of adaptive radiation.

20. Give detailed account on dating methods.

AFF–1082

Sub. Code
4MZO3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

Elective — BIOPHYSICS AND INSTRUMENTATION

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Covalent bond.

2. Hydrophobic interactions.

3. Secondary structure of protein.

4. Polymorphism.

5. GM counter.

6. Isotopes.

7. HPLC.

8. PAGE.

9. XRD.

10. SEM.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain ionic bond with example.

Or

(b) Describe hydrogen bond.

12. (a) Explain primary and secondary structure of proteins.

Or

(b) Write short notes on structural polymorphism of DNA.

13. (a) Describe the structure and function of Scintillation counter.

Or

(b) Write down the safety measures to be followed in handling radioactive isotopes.

14. (a) What are the basic principles of sedimentation?

Or

(b) Give a brief account on electrophoresis techniques.

15. (a) Describe TEM.

Or

(b) Write short notes on NMR.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the strong and weak interactions of macromolecules in detail.

17. Classify proteins based on their structure.

18. Write down the applications of radioactive isotopes in biological studies.

19. Give a detailed account on Electrophoretic techniques.

20. Explain the structure function and uses of electron microscopes.

AFF–1083

Sub. Code
4MZO3E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

Elective — ANIMAL CELL CULTURE TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Established cell line culture.

2. Simple growth medium.

3. CO_2 incubator.

4. Minimal media.

5. Cytotoxicity.

6. Cell separation.

7. Micromanipulation.

8. Vaccines.

9. Apoptosis.

10. Histolytic cultures.

Part B (5 × 5 = 25)

Answer **all** questions, choosing either (a) or (b).

11. (a) Write note on animal cell culture equipments.

Or

(b) Briefly describe about the structure and organization of animal cell.

12. (a) Write note on role of serum and supplement in culture medium.

Or

(b) What are the different types of culture medium?

13. (a) What are the methods used for maintaining cell culture?

Or

(b) How will you get the primary cell culture?

14. (a) Write note on cell synchronization.

Or

(b) Briefly describe about the cell culture based vaccines.

15. (a) Write note on cell preservation.

Or

(b) Explain about mass cell cultivation.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Differentiate the primary and established cell line culture.

17. Discuss about chemical and physical function of constituent in culture media.

18. Write a note on *in vitro* cell culture.

19. Explain in detail about embryonic stem cell.

20. What is hybridoma technology and explain its applications?

AFF–1085

Sub. Code
4MZO3E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Zoology

Elective — TRANSGENIC TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Sterile animals.

2. Transgenic species.

3. Purines and Pyrimidines.

4. Cytogenetics.

5. Live stocks.

6. Maintenance of Breeding animals.

7. Ligation and transfection.

8. DNA finger printing.

9. Cell culture.

10. Animal cloning.

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) Write note on artificial insemination.

Or

(b) Briefly describe about the Transgenic species.

12. (a) Chemical nature and functions of nucleic acids.

Or

(b) Write note on immunogenetics.

13. (a) What are the strategies for improvement of livestock for milk?

Or

(b) Describe about disease resistance of breeding animals.

14. (a) Briefly describe about the *In vitro* fertilization.

Or

(b) Write note on super ovulation and collection of eggs.

15. (a) Write the history and importance of cell technology.

Or

(b) What are the uses of cell technology in industry and research?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about the collection and preservation of sperms.

17. Describe about the recombinant DNA technology.

18. What are the strategies used for the improvement of livestock?

19. Explain about the cloning of Dolly.

20. Write note on merits and demerits of cell technology.